



RT6863

APPLICATIONS

High Performance Audio

Portable/Mobile Audio

High Resolution Audio ADC Analog Stage

High Resolution Audio DAC Analog Stage

Low Noise Amplifier Front End

Headphone/Earphone Driver

Audio Active Filters

Audio Preamplifier

Audio Line Receiver/Driver

FEATURES & BENEFINTS

General & DC Performance

Open-Loop Gain : 135dB

Power Supply : $\pm 2.5V$ to $\pm 17V$

Offset Voltage < $\pm 80\mu V$

CMRR : 120dB

PSRR : 135dB

Quiescent Current : 4.2mA / Per Amp

Unity Gain Stable

Gain Bandwidth : 18MHz

Enable Control @ DFN & CSP Package

Dynamic & AC Performance

Audiophile BJT Input Stage

Input Voltage Noise : 1.8nV rt/Hz @ 1kHz

Input Voltage Noise 1/f Corner : 60Hz

0.1Hz - 10Hz Noise : 100nV Peak-To-Peak

Rail-To-Rail Output

Slew Rate : 15V/ μS

Linear Output Current > 100mA

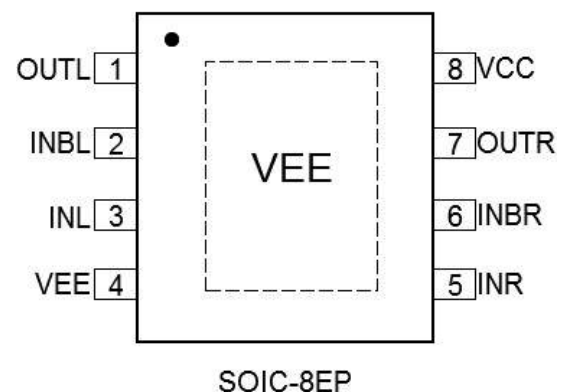
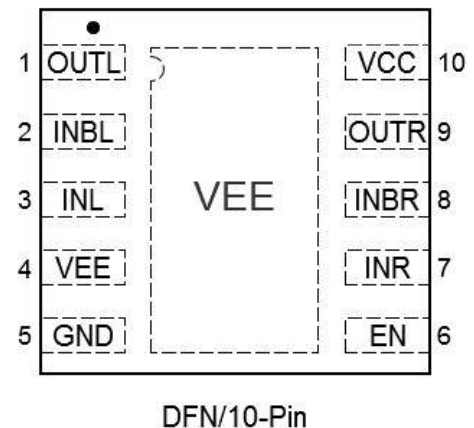
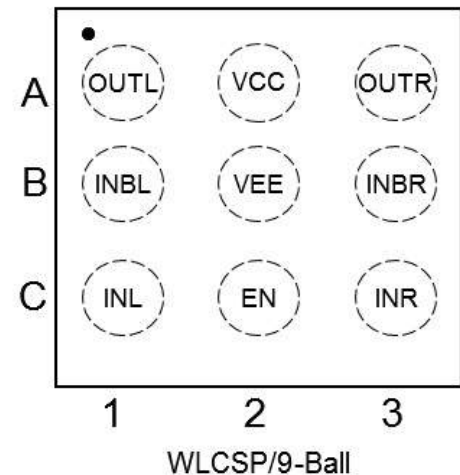
THD+n < -116dB @ 1kHz

IMD < -120dB @ CCIF 19k/20k 1:1

High Quality Audiophile
Operational Amplifier/Driver

PIN CONFIGURATION

(TOP VIEW)





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ELECTRICAL CHARACTERISTICS

Test Conditions (unless otherwise specified)

1. TA=25°C, VCC/VEE = ±2.25V to ±17V, R_L = 100kΩ.

2. Audio Precision SYS-2722.

AUDIO PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
Signal-to-Noise Ratio	SNR	VOUT=2V rms , A-weighted		129		dB
Total Harmonic Distortion Plus Noise	THD+N	VOUT=3.16V rms		-116		dB
		POUT=120mW, RL=32Ω		-114		dB
Inter-Modulation Distortion	IMD	VOUT=2V rms CCIF IMD 19k/20k 1:1		-120		dB
Inter-Modulation Distortion	IMD	VOUT=2V rms SMPTE/DIN IMD 70/7k 4:1		-106		dB
Inter-Modulation Distortion	DIM/TIM	VOUT=2V rms DIM30		-105		dB
EN / Normal	VIH	VEE = Reference "Ground"(RT6863C)	VEE+1.6		VEE+5.5	V
		GND= Reference "Ground"(RT6863D)	1.6		MIN(5.5,+Vs)	V
EN / Shut Down	VIL	VEE = Reference "Ground" (RT6863C)	VEE		VEE+0.6	V
		GND= Reference "Ground"(RT6863D)	-0.3		0.6	V



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PARAMETER	SYMBOL	CONDITIONS	MIN	TYP	MAX	UNITS
INPUT CHARACTERISTICS						
Input Offset Voltage	V_{OS}			± 80		μV
Input Common Mode Voltage Range	V_{CM}		$V_{EE}+2$		$V_{CC}-2$	V
Common Mode Rejection Ratio	CMRR	$(V_{EE})+2V \leq V_{CM} \leq (V_{CC})-2V$		120		dB
Open-Loop Gain	A_{VOL}	$(V_{EE})+0.2V \leq V_{OL} \leq (V_{CC})-0.2V$		135		dB
Input Offset Voltage Drift	$\Delta V_{OS}/\Delta T$			1		$\mu V/^{\circ}C$
OUTPUT CHARACTERISTICS						
Output Voltage Swing from Rail	V_o	$R_L = 10K\Omega$		± 0.2		V
Linear Output Current	I_o			100		mA
FREQUENCY RESPONSE						
Gain-Bandwidth Product	GBP	Gain = 1		18		MHz
Slew Rate	SR			15		V/ μs
Channel Isolation & Separation		$f = 1kHz$		-140		dB
NOISE PERFORMANCE						
Input Voltage Noise Density	e_n	$f = 1kHz$		1.8		$nV/\sqrt{Hz}$
POWER SUPPLY						
Supply Voltage	V_s		± 2.5		± 17	V
Quiescent Current/Amplifier	I_Q	$I_{OUT} = 0A$		4.2		mA
Power Supply Rejection Ratio	PSRR	$V_{CC}/V_{EE} = \pm 2.5V$ to $\pm 17V$		135		dB

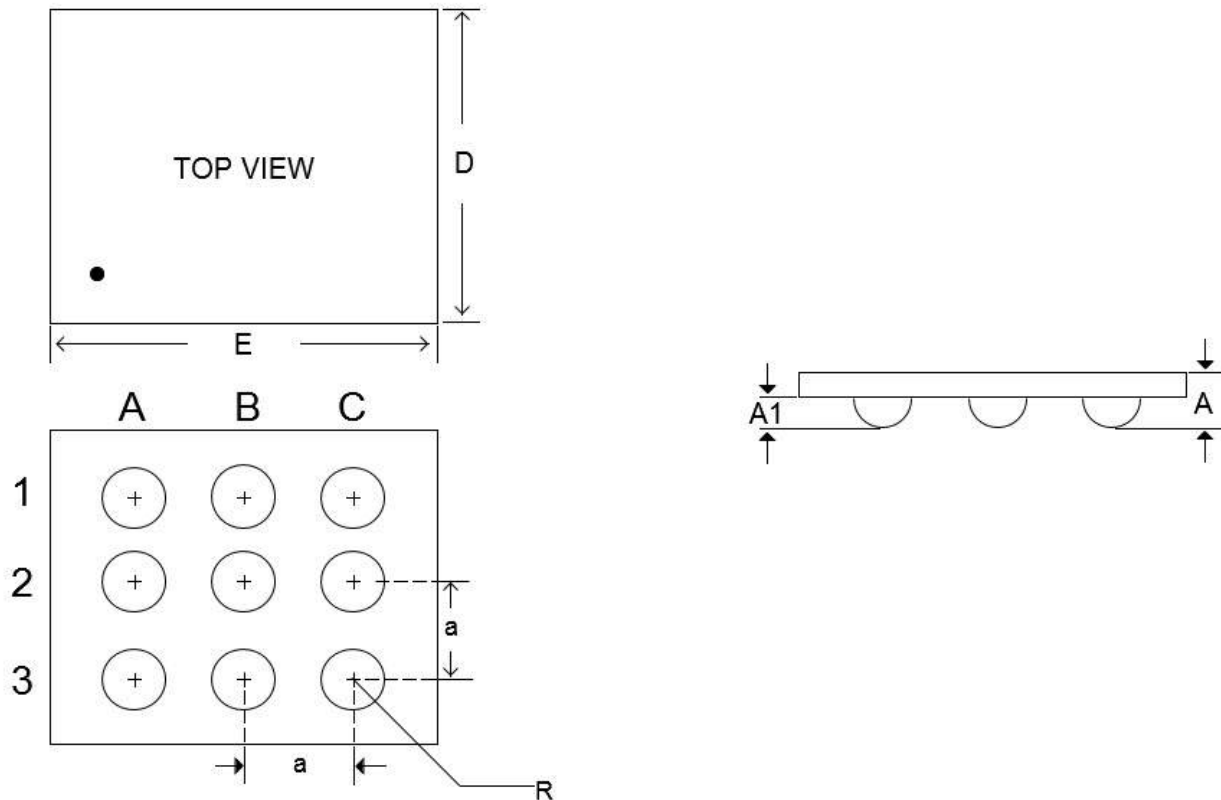
ABSOLUTE MAXIMUM RATINGS

PARAMETER	RATING	UNITS
Storage temperature	-55 to +150	$^{\circ}C$
Junction Temperature	+150	$^{\circ}C$
Power Supply VCC-VEE	36 (+/-18)	V
Input Voltage	$V_{CC}(V_{EE}) \pm 0.3$	V
ESD Protection		
Human Body Model (HBM)	2000	V
Machine Model (MM)	200	

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OUTLINE DIMENSIONS

WLCSP 1.7mm×1.5mm 9-Ball

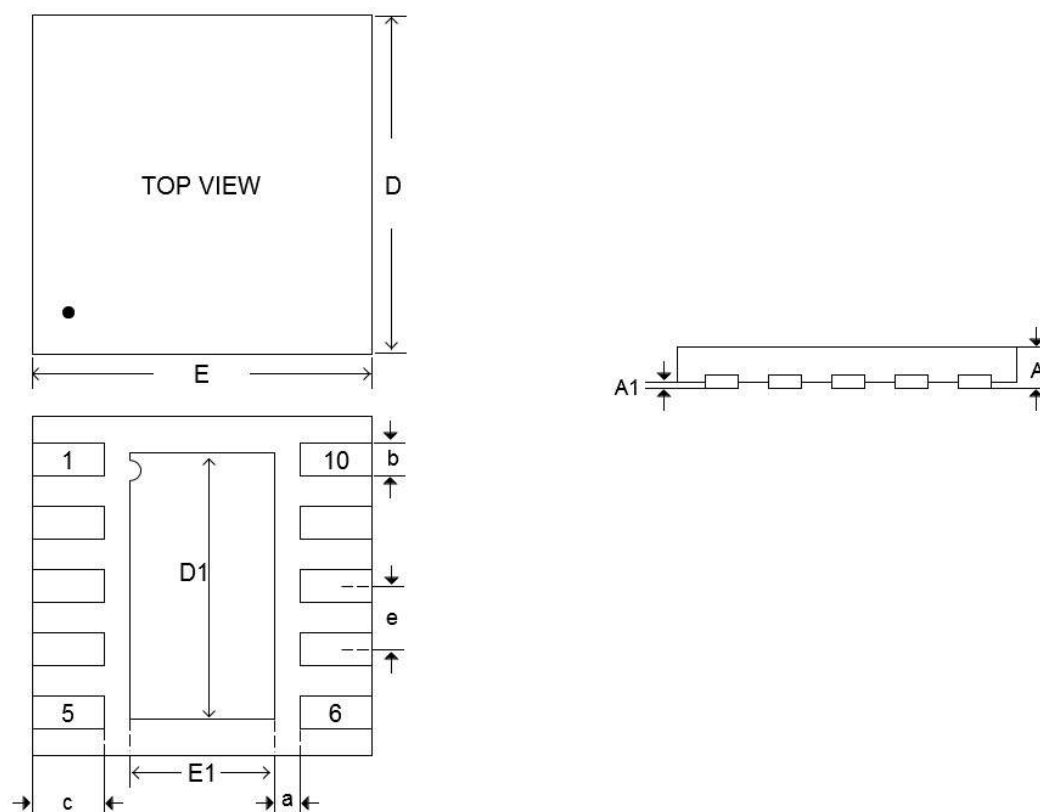


Symbol	Dimensions Millimeters		
	Normal	Min	Max
A	0.575	0.525	0.625
A1	0.194	0.169	0.219
D	1.500	1.485	1.515
E	1.700	1.685	1.715
R	0.268	0.228	0.308
a	0.4	0.4	0.4

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OUTLINE DIMENSIONS

DFN 3mm×3mm 10-Pin

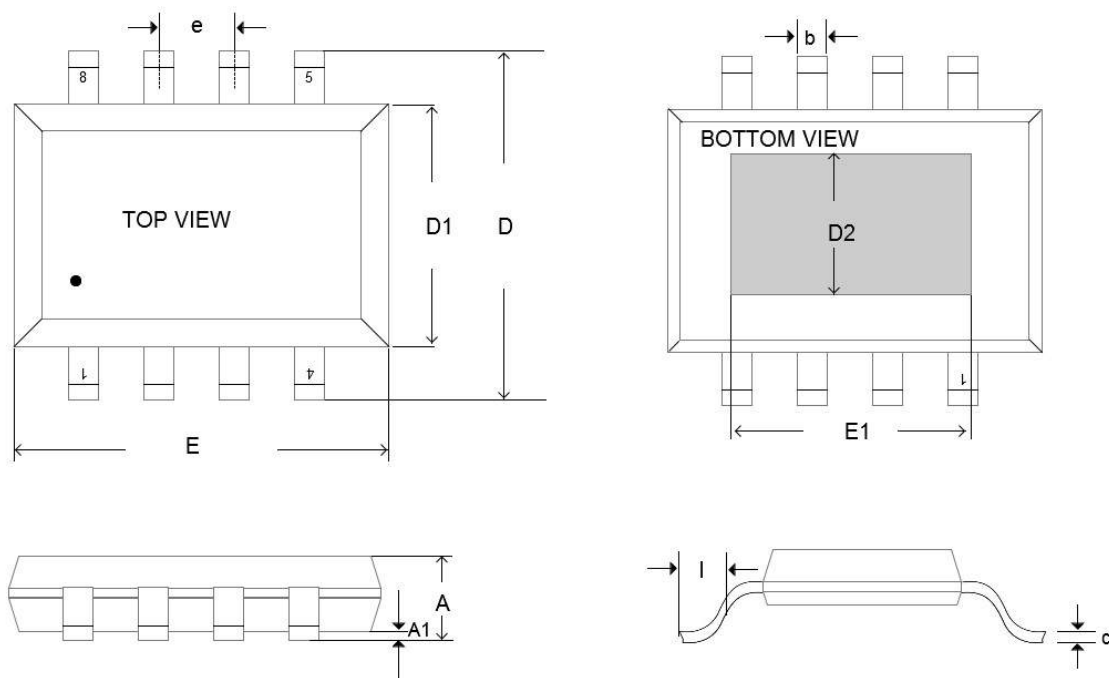


Symbol	Dimensions In Millimeters		
	Normal	Min	Max
A	0.750	0.700	0.800
A1	0.025	0.000	0.050
D	3.000	2.900	3.100
D1	2.400	2.300	2.600
E	3.000	2.900	3.100
E1	1.700	1.500	1.800
a	0.225	0.200	0.250
b	0.200	0.180	0.300
c	0.400	0.300	0.500
e	0.500 TYP		

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OUTLINE DIMENSIONS

SOIC-8EP



Symbol	Dimensions In Millimeters		
	Normal	Min	Max
A	1.650	1.600	1.700
A1	0.050	0.000	0.100
D	6.000	5.800	6.200
D1	3.900	3.800	4.000
D2	2.413	2.313	2.513
E	4.900	4.700	5.100
E1	3.302	3.202	3.402
b	0.355	0.330	0.510
c	0.200	0.170	0.250
e	1.270	1.270	1.270
I	0.820	0.400	1.270



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ORDERING INFORMATION

Part Number	Description	Package
RT6863S	Audio Operational Amplifier/Driver	SOIC-8EP
RT6863D	Audio Operational Amplifier/Driver	DFN 3mmx3mm 10-Pin
RT6863C	Audio Operational Amplifier/Driver	WLCSP 1.7mmx1.5mm 9-Ball

REVISION HISTORY

Revision	Date	Notes
A	AUG2015	Initial Version
B	SEP2015	General Update
C	DEC2015	Enable Control Update
D	21.DEC 2015	Modify c and e Value in DFN-10P
E	28.DEC 2015	Enable Control Update(RT6863D)